

Conditional Rate–Distortion Benefits of 3D DT-CWT Preprocessing for Noisy Low-Bitrate H.264 Video

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Abstract—This study evaluates whether three-dimensional dual-tree complex wavelet transform (3D DT-CWT) preprocessing is beneficial before low-bitrate H.264/x264 encoding. Rather than assuming a universal gain, we test a conditional hypothesis: preprocessing should help noisy inputs, especially at low bitrate, while potentially harming already-clean inputs. Experiments were conducted on the *akiyo* and *foreman* CIF sequences with additive Gaussian noise levels $\sigma \in \{0, 5, 10, 15\}$ and target bitrates from 100 to 500 kbps. The proposed method was compared with direct x264 encoding and additional denoising baselines. Results show a clear condition-dependent pattern. On clean inputs, proposed preprocessing slightly reduced mean PSNR by 0.15–0.24 dB. On noisy inputs, gains increased with noise level: up to 2.82 dB on *akiyo* and 0.92 dB on *foreman*. Reliable metrics such as PSNR-B and edge PSNR follow the same broad trend. However, the pre/post analysis indicates that post-x264 gains are smaller than pre-x264 denoising gains, so the present evidence supports robustness of the preprocessing benefit through encoding rather than amplification by the codec itself.

Index Terms—video coding, denoising, DT-CWT, H.264, preprocessing, rate–distortion

I. INTRODUCTION

Low-bitrate video coding is especially sensitive to sensor noise because noisy high-frequency components consume bitrate while providing little perceptual value. A common response is to denoise before encoding, but preprocessing is not free: if the input is already clean, filtering may remove useful detail and reduce objective quality. This makes the value of preprocessing strongly conditional on the input regime.

The dual-tree complex wavelet transform (DT-CWT) is attractive for denoising because it offers improved shift invariance and directional selectivity compared with separable real wavelets [1], [2]. Extending the transform to three dimensions allows spatial and temporal structure to be treated jointly, which is useful for video denoising. Prior work has studied DT-CWT for restoration and wavelet-domain video coding, but the practical question addressed here is narrower: whether an external 3D DT-CWT preprocessing stage improves the rate–distortion behavior of a standard x264 pipeline under noisy low-bitrate conditions.

We make three contributions:

- We evaluate 3D DT-CWT preprocessing under both clean and noisy conditions instead of reporting only favorable noisy cases.

- We compare post-encoding gains with pre-encoding denoising gains to separate pure restoration benefit from codec interaction.
- We report multiple objective metrics, emphasizing PSNR, PSNR-B, and edge PSNR rather than relying on one headline score.

II. METHOD

A. Pipeline

Each Y4M input sequence is split into luminance and chrominance planes. The proposed pipeline applies 3D DT-CWT shrinkage to the luminance channel, keeps chroma unchanged in the primary experiments, then encodes the result with x264. Chunks are processed with overlap-save handling to reduce temporal boundary artifacts. The implementation automatically uses a CUDA path when available and otherwise falls back to the CPU reference implementation.

B. Thresholding

Let Y_h denote high-pass DT-CWT coefficients. The proposed processor estimates noise from the finest-scale real coefficients using a median absolute deviation estimate and applies BayesShrink-style soft thresholding per subband:

$$T = \frac{\sigma_n^2}{\sigma_x} \alpha_l, \quad (1)$$

where σ_n^2 is estimated noise variance, σ_x is estimated signal standard deviation, and α_l scales the threshold by transform level. This study uses the adaptive mode from the implementation, with a base threshold of 0.03.

C. Pre/Post Analysis

To determine whether encoding itself amplifies the benefit of preprocessing, we evaluate:

$$\Delta_{\text{pre}} = Q(\text{preprocessed}) - Q(\text{input}), \quad (2)$$

$$\Delta_{\text{post}} = Q(\text{preprocessed}+\text{x264}) - Q(\text{baseline}+\text{x264}), \quad (3)$$

and define

$$\text{codec gain} = \Delta_{\text{post}} - \Delta_{\text{pre}}. \quad (4)$$

Positive codec gain would indicate that the codec magnifies a preprocessing benefit; negative values indicate that part of the preprocessing gain is lost after encoding.

TABLE I
PROPOSED MINUS BASELINE MEAN PSNR DIFFERENCE (DB).

Sequence	$\sigma = 0$	$\sigma = 5$	$\sigma = 10$	$\sigma = 15$
<i>akiyo</i>	-0.24	1.34	2.05	2.82
<i>foreman</i>	-0.15	0.02	0.45	0.92

III. EXPERIMENTAL SETUP

A. Sequences and Conditions

Experiments used two CIF sequences: *akiyo* and *foreman*. Gaussian noise was added to the luminance channel with $\sigma \in \{0, 5, 10, 15\}$. Five target bitrates were tested: 100, 200, 300, 400, and 500 kbps.

B. Compared Methods

The evaluated methods were direct x264 encoding, x264 noise reduction, FFmpeg hqdn3d, 3D DWT preprocessing, Gaussian blur preprocessing, and the proposed 3D DT-CWT preprocessing. The main analysis below focuses on proposed versus direct x264 because that pair directly tests the research hypothesis; the complete long-form outputs are preserved in the experiment CSV files.

C. Metrics

The primary metrics are PSNR-Y, PSNR-B, and edge PSNR (EPSNR). SSIM, GBIM, and MEPR were also logged. MS-SSIM and VMAF were unavailable in the current output set because the executed FFmpeg build did not produce valid libvmaf measurements for these runs, so they are not used for conclusions.

IV. RESULTS

A. Condition-Dependent PSNR

Table I summarizes the mean PSNR difference of proposed preprocessing versus direct x264. Clean inputs show slight degradation, while noisy inputs show improvement that increases with σ . The effect is strongest on *akiyo*; *foreman* requires stronger noise before the advantage becomes pronounced.

B. Reliable Auxiliary Metrics

Table II reports mean post-encoding gains in PSNR-B and EPSNR. The same broad pattern appears: clean content is slightly harmed, whereas higher-noise conditions generally benefit from preprocessing. At $\sigma = 5$ on *foreman*, the PSNR gain is marginal and auxiliary metrics remain slightly negative, suggesting that the operating point is close to a crossover regime rather than a robust win.

C. BD-Rate and Pre/Post Behavior

Figure 3 visualizes BD-rate where the curves provide sufficient overlap for reliable estimation. Clean inputs produce positive BD-rate values, consistent with the small PSNR loss. Several noisy conditions produce strongly negative BD-rate values, indicating bitrate savings at matched quality. However,

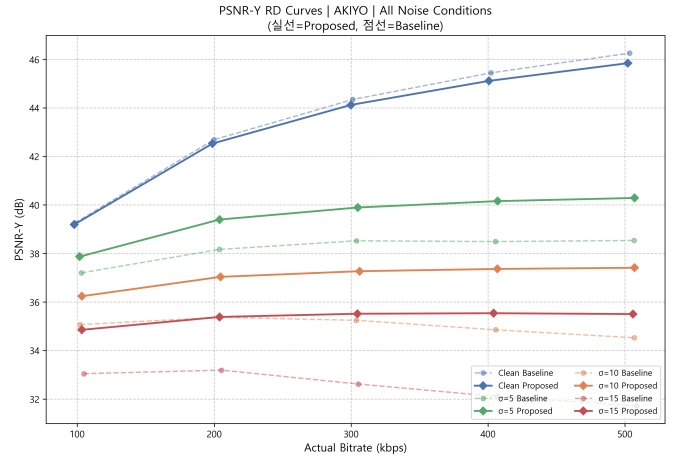


Fig. 1. PSNR RD curves for *akiyo*. Clean inputs favor the baseline, while noisy inputs favor the proposed preprocessing path.

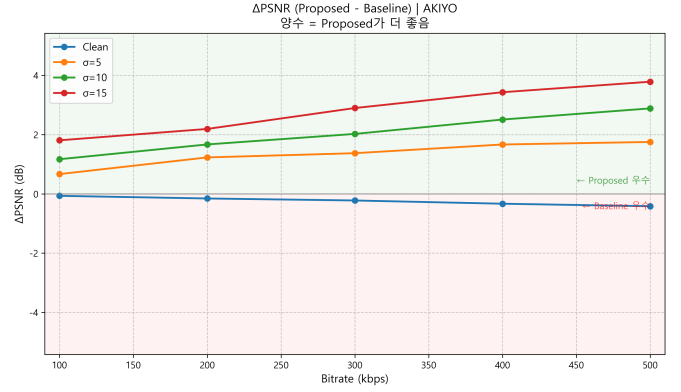


Fig. 2. Per-bitrate PSNR delta for *akiyo*. The sign reversal between clean and noisy conditions is the central empirical finding.

some *akiyo* high-noise conditions yield undefined BD-rate because the RD curves do not have enough overlapping quality range; in those cases, mean and low-bitrate deltas are more interpretable than extrapolated cubic fits.

The pre/post study does *not* support the stronger claim that x264 amplifies the denoising benefit. For the proposed method, measured codec gain in PSNR is negative for all noisy conditions with valid pre-analysis: -3.10 dB for *akiyo* at $\sigma = 5$, -4.74 dB at $\sigma = 10$, -5.23 dB at $\sigma = 15$, and between -1.93 and -4.58 dB for *foreman*. Figure 4 summarizes this behavior. The defensible conclusion is therefore that 3D DT-CWT preprocessing remains beneficial after low-bitrate encoding under noisy conditions, but the current experiment does not show codec-side amplification.

V. DISCUSSION

The results support a conditional deployment rule rather than unconditional preprocessing. If the source is clean, the proposed filter slightly degrades rate-distortion performance. If the source is visibly noisy, especially for low-motion content

TABLE II
MEAN POST-ENCODING GAINS OF PROPOSED PREPROCESSING OVER
BASELINE (dB).

Sequence	Metric	$\sigma = 0$	$\sigma = 10$	$\sigma = 15$
<i>akiyo</i>	PSNR-B	-0.11	2.44	3.67
	EPSNR	-0.11	1.22	1.39
<i>foreman</i>	PSNR-B	-0.08	0.61	1.75
	EPSNR	-0.11	0.07	0.26

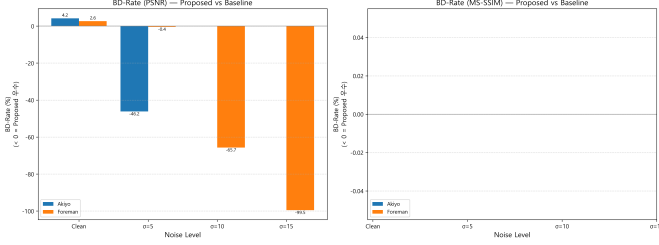


Fig. 3. BD-rate comparison across conditions. Negative values favor the proposed method.

such as *akiyo*, the same preprocessing becomes clearly advantageous. This suggests that a practical system should estimate input noise and enable the filter selectively.

The content dependence is also important. *foreman* contains more motion and structural detail than *akiyo*, and the gains emerge more gradually. This is consistent with the risk that denoising may suppress useful detail or motion cues when noise is not dominant.

Several limitations remain. Only two sequences are represented in the current completed output set, the sample size is too small for broad generalization, and the missing MS-SSIM/VMAF values limit perceptual interpretation. In addition, BD-rate can become undefined when noisy and denoised curves do not sufficiently overlap. Future work should extend the experiment to more sequences, verify `libvmaf`-based metrics, and evaluate a learned or rule-based gating policy that activates preprocessing only when estimated noise exceeds a threshold.

VI. CONCLUSION

3D DT-CWT preprocessing is not universally beneficial before x264 encoding, but it yields clear gains for noisy low-bitrate video. Across the current experiments, clean inputs incurred small PSNR losses, while noisy inputs gained up to 2.82 dB mean PSNR and substantial low-bitrate improvements. The evidence supports a conditional preprocessing strategy driven by source noise level. The pre/post ablation further shows that the improvement mainly originates from denoising itself and is preserved, though not amplified, by subsequent encoding.

REFERENCES

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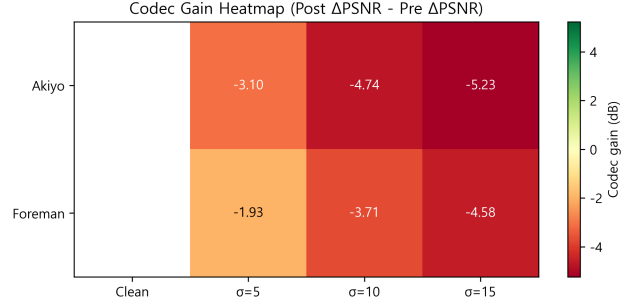


Fig. 4. Codec gain heatmap. Negative values indicate that post-encoding improvement is smaller than pre-encoding denoising improvement.

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